

Lesson 3

The World's Most Bicycle-friendly City

[本文文字起こし](#)・[和訳](#)/[語彙](#)/[Questions](#)・[解答](#)/[Whole Text View](#)/[Grammar](#)・[解答](#)

画像の教科書ページをもとに、学習しやすい順序に再構成しました。本文の後に和訳、各Partの後に欄外語彙と設問・解答を掲載しています。

1. 本文・和訳

Part 1 Copenhagen: The world's most bicycle-friendly city

You are reading an article about a bicycle-friendly city.

【和訳】あなたは、自転車に優しい都市についての記事を読んでいます。

How often do you ride a bicycle? Some people might say every day, others might say just occasionally. In Japan, 13 percent of trips are made on a bicycle. That's a comparatively high percentage among the selected countries below.

【和訳】あなたはどのくらいの頻度で自転車に乗りますか。毎日と答える人もいれば、たまにしか乗らないと答える人もいるでしょう。日本では、移動の13パーセントが自転車で行われています。これは、下に挙げられた国々の中では比較的高い割合です。

Cycling is attracting a lot of public attention these days. That's because it's an eco-friendly form of transportation that emits no greenhouse gases. What's more, it keeps people healthy. It may prevent you from getting diseases such as diabetes and cancer. If you are out of shape, you had better start exercising, and cycling is a good option.

【和訳】近ごろ、自転車に乗ることが世間の大きな注目を集めています。それは、温室効果ガスを排出しない環境に優しい交通手段だからです。さらに、自転車は人々を健康に保ちます。糖尿病やがんなどの病気にかかるのを防いでくれるかもしれません。運動不足なら、運動を始めたほうがよく、自転車はよい選択肢です。

Many European cities have been taking active steps to get people into the habit of cycling. Copenhagen, the capital of Denmark, is a leader among them. It has implemented some unique policies with the goal of becoming the world's best bike city.

【和訳】ヨーロッパの多くの都市は、人々に自転車に乗る習慣を身につけてもらうため、積極的な対策を講じてきました。デンマークの首都コペンハーゲンも、その先頭に立っています。世界一の自転車都市になることを目標に、独自の政策をいくつか実施してきました。

Part 1 New Words

Word / Phrase	和訳
Copenhagen	コペンハーゲン
occasionally	時々、たまに
comparatively	比較的に
eco-friendly	環境に優しい
emit	排出する、放出する
greenhouse	温室（の）
diabetes	糖尿病
Denmark	デンマーク
implement	実施する、実行する

Questions 1-3 (文字起こし・解答)

1. Do people in Japan use bicycles more than people in many other countries?

Answer: Yes, they do. In Japan, 13 percent of trips are made on a bicycle, which is comparatively high.

答えの意味：はい。日本では移動の13%が自転車で行われており、比較的高い割合だからです。

2. Why has cycling been attracting people's attention these days?

Answer: Because it is eco-friendly, emits no greenhouse gases, and helps keep people healthy.

答えの意味：環境に優しく、温室効果ガスを排出せず、人々の健康維持にも役立つからです。

3. What is Copenhagen trying to do now?

Answer: It is trying to become the world's best bike city.

答えの意味：世界一の自転車都市になろうとしています。

Opinion 1 (解答例)

Do you like cycling? If you do, what is good about cycling?

Yes, I do. Cycling is good exercise, and it is an eco-friendly way to travel.

はい。自転車はよい運動になり、環境に優しい移動方法です。

Part 2 The Green Wave

Let's find out what exactly Copenhagen has been doing. To begin with, let's look at the Green Wave. As long as cyclists maintain a speed of 20 kilometers per hour, the traffic lights are green all the way during rush hour on major bicycle routes. The road signals are operated simultaneously with the movements of the cyclists. In the past, there was traffic congestion because people were cycling at different speeds. Now everyone cycles at a speed of 20 kilometers per hour, which is a fairly rapid speed. So there is no traffic congestion.

【和訳】コペンハーゲンが具体的に何をしてきたのか見てみましょう。まず、グリーン・ウェーブを見てみましょう。自転車に乗る人が時速20キロを保つ限り、ラッシュアワーの主要な自転車ルートでは信号がずっと青になります。道路の信号は、自転車に乗る人々の動きと同時に作動するようになっています。以前は、人々が異なる速さで走っていたため渋滞がありました。今では全員が時速20キロというかなり速い速度で走るの、渋滞はありません。

It's true that some cyclists would rather go faster, but they can be a danger to other cyclists, especially during rush hour. The Green Wave makes them travel at a safe speed, because if they go too fast, they'll get caught at a red light. The Green Wave ensures that everyone cycles at the safest speed without sacrificing efficiency.

【和訳】確かに、もっと速く走りたい自転車利用者もありますが、特にラッシュアワーには、ほかの自転車利用者にとって危険になり得ます。グリーン・ウェーブによって、安全な速度で走るようになります。速すぎると赤信号につかまるからです。グリーン・ウェーブは、効率を犠牲にせず、全員が最も安全な速度で走ることを確実にしています。

Part 2 New Words

Word / Phrase	和訳
cyclist	自転車に乗る人、自転車利用者
rush	殺到、混雑；急ぐ
simultaneously	同時に
congestion	混雑、渋滞
fairly	かなり、相当に
sacrifice	犠牲にする
efficiency	効率、能率

Questions 4-5 (文字起こし・解答)

4. How does the Green Wave work?

Answer: If cyclists maintain a speed of 20 kilometers per hour, the traffic lights stay green all the way on major bicycle routes during rush hour.

答えの意味：自転車利用者が時速20キロを保つと、ラッシュアワーの主要ルートで信号がずっと青になります。

5. What happens if someone rides at a faster speed than others during rush hour?

Answer: The cyclist will get caught at a red light and may be a danger to other cyclists.

答えの意味：赤信号につかまり、ほかの自転車利用者にとって危険になる可能性があります。

Opinion 2 (解答例)

What do you think of the Green Wave in Copenhagen?

I think it is a clever system because it makes cycling both safe and efficient.

自転車での移動を安全かつ効率的にする、賢い仕組みだと思います。

Part 3 The Bicycle Snake and Cycle Superhighways

There is also a bicycle bridge spanning the city center called the Bicycle Snake. This bridge serves as a shortcut for cyclists and offers a striking view of the waterfront. It gently snakes along, just as the name suggests. Its gently curving design prevents anyone from going too fast, so everyone can cruise safely.

【和訳】市の中心部に架かる「バイシクル・スネーク」と呼ばれる自転車橋もあります。この橋は自転車利用者の近道となり、水辺の印象的な景色を楽しませてくれます。その名のとおり、緩やかに蛇行しています。穏やかにカーブする設計によって速く走りすぎるのを防ぐため、誰もが安全にゆったり走れます。

For longer journeys, many Cycle Superhighways that connect Copenhagen with surrounding communities have been built. These are designed to encourage suburban residents to commute to work or school by bike. According to residents, they are an absolute pleasure to use.

There are some routes that pass through a forest. "You can enjoy the peacefulness of the forest as you commute," says one user approvingly.

【和訳】より長い移動のためには、コペンハーゲンと周辺地域を結ぶ多くの「サイクル・スーパーハイウェイ」が造られています。これは、郊外の住民に自転車で通勤・通学してもらうために設計されています。住民によれば、利用するのはこの上ない喜びだそうです。森林を通るルートもあります。「通勤しながら森の静けさを楽しめます」と、ある利用者は満足そうに語っています。

All these bike routes have transformed Copenhagen into a cyclist's paradise. No wonder more and more motorists are saying, "I should have left the car at home!"

【和訳】こうした自転車ルートのおかげで、コペンハーゲンを自転車利用者の楽園へと変えました。「車を家に置いてくればよかった」と言う自動車運転者がますます増えているのも不思議ではありません。

Part 3 New Words

Word / Phrase	和訳
span	(橋などが) ～に架かる
shortcut	近道
waterfront	海岸・川岸の地域、水辺
gently	穏やかに、緩やかに
curve	曲がる、カーブする
cruise	ゆったり進む
safely	安全に
superhighway	高速道路
suburban	郊外の
resident	住民
commute	通勤・通学する
absolute	完全な、この上ない
peacefulness	平穏、静けさ
approvingly	満足そうに、賛成して
transform	変える、変形させる
paradise	楽園
motorist	自動車運転者

Questions 6-7 (文字起こし・解答)

6. How does the Bicycle Snake help cyclists?

Answer: It serves as a shortcut, offers a striking waterfront view, and helps cyclists travel safely with its gently curving design.

答えの意味：近道になり、水辺の美しい景色を楽しめ、緩やかなカーブによって安全に走れるようにします。

7. How do residents feel about Cycle Superhighways?

Answer: They feel that the routes are an absolute pleasure to use.

答えの意味：利用するのはこの上ない喜びだと感じています。

Opinion 3 (解答例)

Do you enjoy the view of your surroundings when you come to school?

Yes. I enjoy looking at the trees and the sky on my way to school.

はい。学校へ行く途中で、木々や空を見るのを楽しんでいます。

Part 4 Copenhagenization

Surprisingly, Copenhagen's urban planners once used to favor cars over other forms of transportation. The oil crisis of the 1970s led them to revise their transportation systems so that they would not have to rely on oil. Since then, the city has been installing cycling infrastructure such as the Green Wave and the Bicycle Snake. As a result, it's now quicker to go downtown by bicycle than by car. That has led more people to cycle instead of driving.

【和訳】驚くことに、コペンハーゲンの都市計画者たちは、かつてほかの交通手段よりも自動車を優先していました。1970年代の石油危機をきっかけに、石油に頼らずに済むよう交通システムを見直しました。それ以来、市はグリーン・ウェーブやバイシクル・スネークのような自転車用インフラを整備し続けています。その結果、今では自動車より自転車で中心街へ行くほうが速くなりました。それによって、運転する代わりに自転車に乗る人が増えました。

Making cities healthier and more attractive by adopting bike-friendly policies has come to be known as "Copenhagenization." Even New York is following Copenhagen's example. It has started locating car parking spaces in the middle of the street. That approach is called the "Protected Bicycle Lane" policy and it prevents vehicles from entering or parking in the bike lane. Consequently, bicycle traffic has increased 1.6 times. Moreover, there are fewer accidents. Many other cities may well "Copenhagenize" as they seek to build more sustainable, livable communities. The Copenhagen model is changing the world for the better.

【和訳】自転車に優しい政策を採用して、都市をより健康的で魅力的にすることは、「コペンハーゲン化」として知られるようになりました。ニューヨークでさえコペンハーゲンの例にならっています。ニューヨークは、道路の中央に自動車の駐車スペースを設け始めました。この方式は「保護された自転車レーン」政策と呼ばれ、車両が自転車レーンに進入したり駐車したりするのを防ぎます。その結果、自転車交通量は1.6倍になりました。さらに、事故も減りました。より持続可能で暮らしやすい地域社会を築こうとする中で、ほかの多くの都市も「コペンハーゲン化」するのはもったもです。コペンハーゲン・モデルは世界をよりよい方向へ変えています。

Part 4 New Words

Word / Phrase	和訳
surprisingly	驚くことに
planner	計画立案者、都市計画者
crisis	危機
revise	見直す、改訂する
install	設置する、導入する
infrastructure	インフラ、基盤設備
downtown	中心街へ／中心街の
attractive	魅力的な
adopt	採用する
Copenhagenization	コペンハーゲン化
consequently	その結果、したがって
Copenhagenize	コペンハーゲン化する
sustainable	持続可能な
livable	住みやすい

Questions 8-9 (文字起こし・解答)

8. What happened to the traffic after Copenhagen installed cycling infrastructure?

Answer: It became quicker to go downtown by bicycle than by car, so more people began cycling instead of driving.

答えの意味：中心街へは車より自転車で行くほうが速くなり、運転する代わりに自転車に乗る人が増えました。

9. How has the new policy changed New York?

Answer: Bicycle traffic has increased 1.6 times, and there are fewer accidents.

答えの意味：自転車交通量が1.6倍に増え、事故が減りました。

Opinion 4 (解答例)

Do you think your city can become "Copenhagenized"?

Yes. If the city builds more protected bicycle lanes and safe parking areas, more people will use bicycles.

はい。保護された自転車レーンと安全な駐輪場を増やせば、自転車を使う人が増えると思います。

2. 本文下の重要熟語・和訳

本文各ページの下に太字で掲載されている表現です。意味と教科書の例文もまとめました。

out of shape

意味：運動不足で、体調がよくなって

Example: I haven't been exercising lately, so I'm really out of shape.

例文訳：私は最近運動していないので、本当に運動不足です。

to begin with

意味：まず初めに、第一に

Example: To begin with, let's listen to today's dialog.

例文訳：まず初めに、今日の対話を聞きましょう。

as long as ...

意味：～する限り、～でありさえすれば

Example: As long as you are careful, you'll be fine.

例文訳：注意している限り、大丈夫です。

no wonder ...

意味：～なのも不思議ではない、～なのも当然だ

Example: No wonder she doesn't like crowded places.

例文訳：彼女が混雑した場所を好きではないのも不思議ではありません。

3. Whole Text View (文字起こし)

※教科書のWhole Text Viewは、Part

1～4の本文をまとめて再掲したページです。以下に英文をPart別にまとめます。

Part 1

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How often do you ride a bicycle? Some people might say every day, others might say just occasionally. In Japan, 13 percent of trips are made on a bicycle. That's a comparatively high percentage among the selected countries below.

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Many European cities have been taking active steps to get people into the habit of cycling. Copenhagen, the capital of Denmark, is a leader among them. It has implemented some unique policies with the goal of becoming the world's best bike city.

Part 2

Let's find out what exactly Copenhagen has been doing. To begin with, let's look at the Green Wave. As long as cyclists maintain a speed of 20 kilometers per hour, the traffic lights are green all the way during rush hour on major bicycle routes. The road signals are operated simultaneously with the movements of the cyclists. In the past, there was traffic congestion because people were cycling at different speeds. Now everyone cycles at a speed of 20 kilometers per hour, which is a fairly rapid speed. So there is no traffic congestion.

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Part 3

There is also a bicycle bridge spanning the city center called the Bicycle Snake. This bridge serves as a shortcut for cyclists and offers a striking view of the waterfront. It gently snakes along, just as the name suggests. Its gently curving design prevents anyone from going too fast, so everyone can cruise safely.

For longer journeys, many Cycle Superhighways that connect Copenhagen with surrounding communities have been built. These are designed to encourage suburban residents to commute to work or school by bike. According to residents, they are an absolute pleasure to use. There are some routes that pass through a forest. "You can enjoy the peacefulness of the forest as you commute," says one user approvingly.

All these bike routes have transformed Copenhagen into a cyclist's paradise. No wonder more and more motorists are saying, "I should have left the car at home!"

Part 4

Surprisingly, Copenhagen's urban planners once used to favor cars over other forms of transportation. The oil crisis of the 1970s led them to revise their transportation systems so that they would not have to rely on oil. Since then, the city has been installing cycling infrastructure such as the Green Wave and the Bicycle Snake. As a result, it's now quicker to go downtown by bicycle than by car. That has led more people to cycle instead of driving.

Making cities healthier and more attractive by adopting bike-friendly policies has come to be known as "Copenhagenization." Even New York is following Copenhagen's example. It has started locating car parking spaces in the middle of the street. That approach is called the "Protected Bicycle Lane" policy and it prevents vehicles from entering or parking in the bike lane. Consequently, bicycle traffic has increased 1.6 times. Moreover, there are fewer accidents. Many other cities may well "Copenhagenize" as they seek to build more sustainable, livable communities. The Copenhagen model is changing the world for the better.

Comprehension 1 Give a title to each part.

- (a) Helpful signals: The Green Wave
- (b) Copenhagen: A bicycle-friendly city
- (c) Making a city Copenhagenized
- (d) Bicycle Snake and Cycle Superhighways

【答え】 Part 1 = (b) / Part 2 = (a) / Part 3 = (d) / Part 4 = (c)

Comprehension 2: Listen to the sentences and circle T if the sentence is true. Circle F if the sentence is false. (a) T / F (b) T / F (c) T / F (d) T / F

※音声文が写真内に掲載されていないため、T/Fの判定はできません。

4. Grammar (文字起こし・解答)

G1 助動詞 + have + 過去分詞

過去のことについての推量や、過去のことについての非難・後悔を表す。

1. I can't find my English textbook. I may have left it at home. (～したかもしれない)
2. Ted hasn't shown up yet. He must have forgotten our appointment. (～したにちがいない)
3. Susan cannot have told a lie. She is an honest person. (～したはずがない)
4. No wonder more and more motorists are saying, "I should have left the car at home!" (～すべきだったのに <しなかった>)

G2 助動詞に関連する表現

1. If you are out of shape, you had better start exercising. (～したほうがよい)
2. It's true that some cyclists would rather go faster. (むしろ～したい)
3. Copenhagen's urban planners once used to favor cars over other forms of transportation. (以前は～だった)
4. Many other cities may well "Copenhagenize." (～するのももっともだ)

Exercises 1

Fill in the blanks and complete the conversations. Use cannot, must, or should.

1. A: The cake in the fridge is gone!
B: Bill () () () it. I'm sure of it! [eat]

【答え】 Bill must have eaten it.

2. A: Betty has not come yet. I wonder if she got lost.
B: She () () () lost. She has come here many times so far. [get]

【答え】 She cannot have gotten lost. (British Englishでは got も用いられます。)

3. (Talk about something you regret not doing in the past by following the example.)

Example: I lost a tennis match last week. I couldn't practice enough because I was busy studying for the exam. I should have practiced more.

【解答例】 I failed a test last month. I didn't study enough because I was busy. I should have studied harder.

Exercises 2

Fill in the blanks and complete the conversations.

1. A: Laura looks mad at Jake.
B: She _____ at him. He is always skipping brass band practice. (怒るのももっともだ)

【答え】 She may well be mad at him.

2. A: Sue, let's go to the new Italian restaurant downtown.

B: Maybe another time. I _____ dinner at home. (むしろ家で食べたい)

【答え】 I would rather eat dinner at home.

3. (Talk about what you used to do in the past and what you often do now by following the example.)

Example: I used to read a lot of fantasy novels when I was in elementary school, but now I read a lot of mystery novels.

【解答例】 I used to play video games after school, but now I often study English.

5. 最後のページの単語リスト・和訳

Word / Phrase	和訳
destination	目的地
compact	小型の、コンパクトな
inch	インチ
freight	貨物
transport	輸送する、運ぶ
load	荷物、積み荷
gravity	重力
maximum	最大の、最大限
navy	濃紺、ネイビー
boost	押し上げるもの、増強
khaki	カーキ色
tandem	二人乗りの、タンデム
output	出力
consultation	相談
height	高さ、身長
length	長さ
dealer	販売店、取扱店
warranty	保証、保証書
discount	割引
frame	フレーム、車体
valid	有効な

参考 : Real Life Information 見出し

Folding Bike: Take your bike to new

destinations! (折りたたみ自転車: 自転車を新しい目的地へ持って行こう!)

Freight Bike: Easy to carry a heavy load! (貨物自転車: 重い荷物も簡単に運べる!)

E-bike: Power assist gives you a boost! (電動自転車: 電動アシストが力を貸してくれる!)

Tandem Bike: Enjoy cycling with your

friends. (タンデム自転車: 友達とサイクリングを楽しもう。)